Atiyah-Macdonald Exercises

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January 16, 2026

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1 Rings and Ideals

1.1 Exercise - Complete

Proof. By assumption, $x^m = 0$ for some m. Then,

$$(1+x)\sum_{i=0}^{m-1}(-1)^i x^i = 1 + (-1)^{m-1}x^m = 1,$$

so $1+x$ is a unit.

Let x be nilpotent and u be a unit. Then, $u + x = u(1 + u^{-1}x)$. But $u^{-1}x$ is nilpotent since x is. By the previous result, $1 + u^{-1}x$ is a unit and hence $u(1 + u^{-1}x) = u + x$ is also a unit. □

1.2 Exercise - Complete

(i) Proof. We will prove this by induction. Suppose f has degree 0 with inverse $g = b_0 + \cdots + b_m x^m$. Then, $1 = fb_0 + \cdots + fb_m x^m$. Therefore, since $f \in A$ and $fb_0 = 1$, f is a unit. The converse is trivial since $f = a_0$, proving the base case.

Now, suppose that for all polynomials of degree $n-1$, the proposition holds. Suppose f has degree n and has an inverse $g = b_0 + \cdots + b_m x^m$. Then,

$$\begin{aligned} 1 &= \left(\sum_{i=0}^{n} a_i x^i\right) \cdot \left(\sum_{j=0}^{m} b_j x^j\right) \\ &= \sum_{k=0}^{n+m} \left(\sum_{i=0}^{k} a_i b_{k-i}\right) x^i. \end{aligned}$$

This implies $a_n b_m = 0$. We'll show by induction on r that $a_n^{r+1} b_{m-r} = 0$, the base case already proven. We assume this is true for all $r < l$. Multiplying by a_n^l,

$$\begin{aligned} a_n^l &= \left(\sum_{i=0}^{n} a_i x^i\right) \cdot \left(\sum_{j=0}^{m} a_n^l b_j x^j\right) \\ &= \left(\sum_{i=0}^{n} a_i x^i\right) \cdot \left(\sum_{j=0}^{m-l} a_n^l b_j x^j\right), \end{aligned}$$

where we used the inductive hypothesis to remove higher terms: if $j > m-l$, then $l-(m-j+1) \geq 0$, so $a_n^l b_j = a_n^{l-(m-j+1)}(a_n^{m-j+1} b_{m-(m-j)}) = 0$. The coefficient of highest degree is $a_n \cdot a_n^l b_{m-l}$, so $a_n^{l+1} b_{m-l} = 0$. By induction, $a_n^{r+1} b_{m-r} = 0$ for all r.

From the expansion of $f \cdot g = 1$, we know that $a_0 b_0 = 1$. Then, using our result,

$$a_n^{m+1} = a_0(a_n^{m+1} b_{m-m}) = a_0 \cdot 0 = 0,$$

so a_n is nilpotent. Then, $-a_n x^n$ is nilpotent. Since f is a unit, by Exercise 1, $f + (-a_n x^n)$ is also a unit. However, it has degree $n-1$. By the inductive hypothesis, a_0 is a unit and a_i is nilpotent for $1 \leq i < n$. We've shown a_n is nilpotent, so this direction is complete.

Conversely, suppose a_0 is a unit and that a_i is nilpotent for $1 \leq i \leq n$. Then, $f - a_n x^n$ has degree $n-1$, so by the inductive hypothesis, $f - a_n x^n$ is a unit since its coefficients have the right properties. However, $a_n x^n$ is nilpotent since a_n is, so by Exercise 1, $(f - a_n x^n) + a_n x^n = f$ is a unit, so both directions are complete. By induction, the proposition follows for f of any degree. □

(ii) Proof. We'll prove this by induction on the degree. The base case is trivial since $f = a_0$ in degree 0. Now, suppose for all degrees less than n, the proposition holds.

If f has degree n and is nilpotent, there is an m such that $f^m = 0$. Expanding, the top term is $a_n^m x^{nm}$, which is zero. Therefore, a_n is nilpotent. However, $g = f + (-a_n x^n)$ is also nilpotent since it is the sum of two nilpotent elements (if a and b are the required nilpotent powers, take $a + b$; this power will annihilate all terms in the binomial expansion of g). g has degree at most $n - 1$, so by the inductive hypothesis $a_0, \ldots, a_{n-1}$ are all nilpotent. Since a_n is nilpotent, this direction is complete.

Conversely, if f has degree n and all coefficients are nilpotent, take $f = g + a_n x^n$, where g has degree at most $n - 1$. By the inductive hypothesis, g is nilpotent, so f is the sum of two nilpotent elements. Thus, f is nilpotent. By induction, the proposition follows for f of any degree. □

(iii) Proof. The reverse implication is by definition. For the forward implication, suppose f is a zero-divisor. The set of all possible g such that $fg = 0$ has minimal degree m since it is non-empty. Choose g of degree m such that $fg = 0$. If the highest terms are $a_n x^n$ and $b_m x^m$, then $a_n b_m x^{n+m} = 0$ by $fg = 0$, so $a_n b_m = 0$. Note that $a_n g$ annihilates the term of degree m and annihilates f, so $a_n g = 0$; otherwise, we would be violating the minimality of m.

We will show by induction that $a_{n-r} g = 0$, the base case just proven. For the inductive hypothesis, suppose $a_{n-r} g = 0$ for all $r < k$. Then,

$$0 = g \sum_{i=0}^{n} a_i x^i = \sum_{i=0}^{n} (g a_i) x^i = \sum_{i=0}^{n-k} (g a_i) x^i.$$

The top term is $b_m a_{n-k} x^{m+n-k}$, so $b_m a_{n-k} = 0$. Thus, $a_{n-k} g$ has degree less than m and annihilates f. Again, to preserve the minimality of m, we have $a_{n-k} g = 0$. By induction, $a_{n-r} g = 0$ for all r.

For any a_i, we have $a_i g = 0$. Thus,

$$0 = a_i \left(b_0 + \cdots + b_m x^m\right) = a_i b_0 + \cdots + a_i b_m x^m.$$

By definition, $b_m \neq 0$, yet $a_i b_m x^m = 0$. So, $a_i b_m = 0$ for all i. Thus,

$$b_m f = b_m \sum_{i=0}^{n} (b_m a_i) x^i = 0,$$

proving the claim. □

Note: *For use in Exercise 3, we just proved: if $fg = 0$, g annihilates every coefficient of f.*

(iv) Proof. Suppose fg is not primitive. Then, the ideal generated by its coefficients lie in a maximal ideal $\mathfrak{m}$. Thus, under the ring homomorphism $A[x] \to (A/\mathfrak{m})[x]$, $0 = \overline{fg} = \overline{f}\,\overline{g}$. If $\overline{f}$ is a zero divisor, then by (iii), there is a nonzero $a \in A/\mathfrak{m}$ such that $a\overline{f} = 0$. But, a is a unit since $A/\mathfrak{m}$ is a field, so $\overline{f} = 0$. The same argument applies to $\overline{g}$. Thus, either $\overline{f}$ or $\overline{g}$ is 0; pulling back whichever is zero, its coefficients must all lie in $\mathfrak{m}$, violating primitivity.

Conversely, suppose f or g is not primitive. Without loss of generality, we can say f is not primitive, so its coefficients lie in a maximal ideal $\mathfrak{m}$. Then,

$$fg = \sum_{k=0}^{n+m} \left(\sum_{i=0}^{k} a_i b_{k-i} \right) x^k.$$

Obviously, $a_i b_{k-i} \in \mathfrak{m}$ since $a_i \in \mathfrak{m}$, so any finite sum of these terms is also. Therefore, the coefficients of fg all lie in $\mathfrak{m}$, so fg is not primitive, concluding the proof. □

1.3 Exercise - Complete

Here, we will frequently use the fact that

$$A[x_1, \ldots, x_n] \cong A[x_1, \ldots, x_{n-1}][x_n],$$

which is easy to see by simply grouping terms on the left-hand side by x_n-degree.

(ii) Proof. We'll prove this by induction, using Exercise 2(ii), which establishes the base case. Now, assume the inductive hypothesis up to $n-1$.

Let $f \in A[x_1, \ldots, x_n]$. We can write $f = f_0 + \cdots + f_m x_n^m \in A[x_1, \ldots, x_{n-1}][x_n]$. Assume f is nilpotent. By Exercise 2(ii), each f_i is nilpotent. Then, by the inductive hypothesis for (ii), all coefficients are nilpotent for each f_i and thus for f. Conversely, if all coefficients of f are nilpotent, then all coefficients of each f_i are nilpotent. By the inductive hypothesis for (ii), each f_i is nilpotent. By Exercise 2(ii), f is nilpotent. By induction, the result follows. □

(i) Proof. We'll prove this by induction, using Exercise 2(i), which establishes the base case. Now, assume the inductive hypothesis up to $n-1$.

Suppose f is a unit in $A[x_1, \ldots, x_n] = A[x_1, \ldots, x_{n-1}][x_n]$. Then, $f = f_0 + f_1 x_n + \cdots + f_m x_n^m$, where $f_i \in A[x_1, \ldots, x_{n-1}]$. By the inductive hypothesis for (i), f_0 is a unit and f_i is nilpotent for $i > 0$. By the inductive hypothesis for (ii), all the coefficients of f_i are nilpotent for $i > 0$. By the inductive hypothesis for f_0, the constant coefficient is a unit, and all other coefficients are nilpotent. Thus, viewing f in $A[x_1, \ldots, x_n]$, the constant term is a unit and all other coefficients are nilpotent.

Conversely, suppose f in $A[x_1, \ldots, x_n]$ with the constant term a unit and all other coefficients nilpotent. Then, we can group terms so that $f = f_0 + f_1 x_n + \cdots + f_m x_n^m$, viewing $f \in A[x_1, \ldots, x_{n-1}][x_n]$. Then, f_0 has a unit for a constant term, with all other coefficients being nilpotent. Thus, f_0 is a unit by the inductive hypothesis for (i). Likewise, f_i for $i > 0$ has all nilpotent coefficients, so Exercise 3(ii) shows that f_i is nilpotent. By Exercise 2(i), f is a unit. By induction, (i) is proven in the general case. □

(iii) Proof. First, we start with a lemma: we want to show by induction that for $f, g \in A[x_1, \ldots, x_n]$, $fg = 0$, we have that g annihilates every coefficient of f. The base case is shown in Exercise 2(iii).

Suppose f is a zero-divisor and $f \in A[x_1, \ldots, x_n] \cong A[x_1, \ldots, x_{n-1}][x_n]$. The set of all possible g such that $fg = 0$ has minimal degree m in x_n since it is non-empty. Choose g of degree m in x_n such that $fg = 0$. If the highest terms are $f_k x_n^k$ and $g_l x_n^l$, then $f_k g_l x^{k+l} = 0$ by $fg = 0$, so $f_k g_l = 0$. Note that $f_k g$ annihilates the term of degree m and annihilates f, so $f_k g = 0$; otherwise, we would be violating the minimality of m.

We will show by induction that $f_{k-r} g = 0$, the base case just proven. For the inductive hypothesis, suppose $f_{k-r} g = 0$ for all $r < q$. Then,

$$0 = g \sum_{i=0}^{k} f_i x_n^i = \sum_{i=0}^{k} (g f_i) x_n^i = \sum_{i=0}^{k-q} (g f_i) x_n^i.$$

The top term is $g_l f_{k-q} x_n^{l+k-q}$, so $g_l f_{k-q} = 0$. Thus, $f_{k-q} g$ has degree less than m and annihilates f. Again, to preserve the minimality of m, we have $f_{k-q} g = 0$. By induction, $f_{k-r} g = 0$ for all r.

For any f_i, we have $f_i g = 0$. Thus,

$$0 = f_i \left(g_0 + \cdots + g_m x_n^m\right) = f_i g_0 + \cdots + f_i g_m x_n^m.$$

For any g_j, we have $f_i g_j x_n^j = 0$. So, $f_i g_j = 0$ for all i and j. By induction, since $f_i, g_j \in A[x_1, \ldots, x_{n_1}]$, g_j annihilates every coefficient of f_i. Since this is true for all j, g annihilates every coefficient of f_i for all i. By induction, for all n, if $f, g \in A[x_1, \ldots, x_n]$ and $fg = 0$, then g annihilates every coefficient of f individually. This proves the lemma.

With this lemma, we can prove (iii). The reverse implication is by definition. For the forward implication, we prove (iii) by induction, using the base case established by Exercise 2(iii). Assume the inductive hypothesis up to $n-1$ and that $f, g \in A[x_1, \ldots, x_n]$, $fg = 0$. By our lemma, and since f and g are symmetrically defined, f annihilates every coefficient of g individually; since g is nonzero, it has some nonzero coefficient $g_\beta \in A$. Thus, $g_\beta f = 0$. By induction, if for any n, $f, g \in A[x_1, \ldots, x_n]$ and $fg = 0$, then $af = 0$ for some nonzero $a \in A$. □

(iv) Proof. Suppose fg is not primitive. Then, the ideal generated by its coefficients lie in a maximal ideal $\mathfrak{m}$. Thus, under the ring homomorphism $A[x_1, \ldots, x_n] \to (A/\mathfrak{m})[x_1, \ldots, x_n]$, $0 = \overline{fg} = \overline{f} \cdot \overline{g}$. If $\overline{f}$ is a zero divisor, then by (iii), there is a nonzero $a \in A/\mathfrak{m}$ such that $a\overline{f} = 0$. But, a is a unit since $A/\mathfrak{m}$ is a field, so $\overline{f} = 0$. The same argument applies to $\overline{g}$. Thus, either $\overline{f}$ or $\overline{g}$ is 0; pulling back whichever is zero, its coefficients must all lie in $\mathfrak{m}$, violating primitivity.

Conversely, suppose f or g is not primitive. Without loss of generality, we can say f is not primitive, so its coefficients lie in a maximal ideal $\mathfrak{m}$. Then, consider the same map as before, $A[x_1, \ldots, x_n] \to (A/\mathfrak{m})[x_1, \ldots, x_n]$. Then, $\overline{fg} = \overline{f} \cdot \overline{g} = 0 \cdot \overline{g} = 0$. In particular, all the coefficients of fg lie in $\mathfrak{m}$, so fg is not primitive, concluding the proof. □

1.4 Exercise - Complete

Proof. Since all maximal ideals are prime ideals, the nilradical (intersection of all prime ideals) is contained in the Jacobson radical (intersection of all maximal ideals) for any ring. Thus, we need to show that any $f(x)$ in the Jacobson radical is nilpotent.

By Proposition 1.9, if $f(x)$ is in the Jacobson radical, then $1 + xf(x)$ is a unit in $A[x]$. We can write this as

$$1 + xf(x) = 1 + a_0 x + \cdots + a_n x^{n+1}.$$

By Exercise 2(i), $a_0, \ldots, a_n$ are nilpotent. Thus, by Exercise 2(ii), f is nilpotent. □

1.5 Exercise

1.6 Exercise - Complete

Proof. Since all maximal ideals are prime, we know that $\mathfrak{N} \subset \mathfrak{R}$. Suppose to the contrary that the ideal $\mathfrak{R}$ is not contained in $\mathfrak{N}$. Then, by assumption, there is some nonzero idempotent $e \in \mathfrak{R}$. Since $e^2 = e$, a simple inductive argument reveals that e cannot be nilpotent.

Let $\mathfrak{p}$ be any prime ideal. Consider the map $A \to A/\mathfrak{p}$. Because $A/\mathfrak{p}$ is an integral domain and $0 = \overline{e^2 - e} = \overline{e}(\overline{e} - 1)$, we have $e \in \mathfrak{p}$ or $e \in 1 + \mathfrak{p}$. If $e \in \mathfrak{p}$ for all prime ideals, then e must also be in nilradical, as it is the intersection of prime ideals. Thus, there is some prime ideal $\mathfrak{p}'$ such that $e \in 1 + \mathfrak{p}'$.

There is some maximal ideal $\mathfrak{m}'$ containing $\mathfrak{p}'$, so $e \in 1 + \mathfrak{m}'$. However, since $e \in \mathfrak{R}$, which is the intersection of all maximal ideals, $e \in \mathfrak{m}'$ also. Together, these two facts imply $1 \in \mathfrak{m}'$, yielding the desired contradiction. □

1.7 Exercise - Complete

Proof. Let $\mathfrak{p}$ be a prime ideal so that $A/\mathfrak{p}$ is an integral domain. For any $y \in A/\mathfrak{p}$ such that $y \neq 0$, there is some $x \in A$, such that $\overline{x} = y$. By assumption, there is an $n > 1$ such that $x^n = x$, so $y^n = y$. Note that $y^{n-1} - 1 \in A/\mathfrak{p}$ and

$$y(y^{n-1} - 1) = y^n - y = 0.$$

Since $A/\mathfrak{p}$ is an integral domain and $y \neq 0$, $y^{n-1} = 1$. In particular, y is a unit. Thus, all nonzero elements are nonzero, implying that $A/\mathfrak{p}$ is a field and $\mathfrak{p}$ is maximal. □

1.8 Exercise - Complete

Proof. Let $\mathcal{S}$ be the set of prime ideals of $\mathcal{S}$. If $\mathfrak{p} \subset \mathfrak{q}$, we'll define a partial ordering by $\mathfrak{q} \leq \mathfrak{p}$. Now, suppose we have a chain $\mathcal{T} \subset \mathcal{S}$. Let

$$Q = \bigcap_{\mathfrak{q} \in \mathcal{T}} \mathfrak{q},$$

which, as an intersection of ideals, is itself an ideal.

Suppose $a, b \in A$ and $ab \in Q$. If $a \notin Q$, then there is some $\mathfrak{r} \in \mathcal{T}$ such that $a \notin \mathfrak{r}$. Since $a \notin \mathfrak{r}$, for any $\mathfrak{q}$ such that $\mathfrak{q} \subset \mathfrak{r}$, $a \notin \mathfrak{q}$. Since $\mathcal{T}$ is a chain,

$$Q = \bigcap_{\mathfrak{q} \leq \mathfrak{r}} \mathfrak{q} \cap \bigcap_{\mathfrak{q} \geq \mathfrak{r}} \mathfrak{q} = \mathfrak{r} \cap \bigcap_{\mathfrak{q} \geq \mathfrak{r}} \mathfrak{q} = \bigcap_{\mathfrak{q} \geq \mathfrak{r}} \mathfrak{q}.$$

Let $\mathfrak{q} \geq \mathfrak{r}$. Then, since $ab \in Q \subset \mathfrak{q}$ and $\mathfrak{q}$ is prime, we must have $b \in \mathfrak{q}$, as we've already shown a cannot be in $\mathfrak{r}$. Thus, $b \in \mathfrak{q}$ for all $\mathfrak{q} \geq \mathfrak{r}$, so $b \in Q$. Thus, Q is a prime ideal.

For all $\mathfrak{q} \in \mathcal{T}$, $Q \subset \mathfrak{q}$ ($Q \geq \mathfrak{q}$) and is thus a maximal element. Using Zorn's lemma, $\mathcal{S}$ has at least one maximal element. By our notation, these are minimal elements with respect to inclusion since we used $\geq$ for $\subset$. □

1.9 Exercise

1.10 Exercise - Complete

Proof. (i) $\implies$ (ii). Since $\mathfrak{N}$ is the intersection of all prime ideals, $\mathfrak{N}$ is the unique prime ideal. By Theorem 1.3, A must have a maximal ideal, but since every maximal ideal is prime and $\mathfrak{N}$ is the only prime ideal, $\mathfrak{N}$ must be maximal. Hence, it is also the only maximal ideal. By Corollary 1.5, every non-unit is contained in a maximal ideal, so every non-unit must be in $\mathfrak{N}$, hence nilpotent.

(ii) $\implies$ (iii). Consider the ring homomorphism $A \to A/\mathfrak{N}$. Every element $x \in A$ is either a unit or nilpotent; if x is a unit, $\overline{x}$ is a unit, and if x is nilpotent, $x \in \mathfrak{N}$, so $\overline{x} = 0$. Since the projection is surjective, every nonzero element of the ring $A/\mathfrak{N}$ is a unit, i.e. $A/\mathfrak{N}$ is a field.

(iii) $\implies$ (i). Since $A/\mathfrak{N}$ is a field, $\mathfrak{N}$ a maximal ideal. All maximal ideals are prime. However, $\mathfrak{N}$ is also the intersection of all prime ideals. Suppose there is another prime ideal $\mathfrak{p}$, not equal to $\mathfrak{N}$. Thus, there is an element $x \in \mathfrak{p} - \mathfrak{N}$ since $\mathfrak{N} \subset \mathfrak{p}$. Under the map $A \to A/\mathfrak{N}$, $\mathfrak{x}$ is a unit since it is non-zero. There is a y in A such that $xy = 1 + \mathfrak{N}$. But, $xy \in \mathfrak{p}$ since $x \in \mathfrak{p}$, so $\mathfrak{p} + \mathfrak{N} = (1)$ and are thus coprime. But $\mathfrak{N} \subset \mathfrak{p}$, so $\mathfrak{p} = (1)$, a contradiction. □

1.11 Exercise - In Progress (iii)

Proof. For every $x \in A$,

$$\begin{aligned}(1+x)^2 &= 1+x \\ 1+2x+x^2 &= 1+x \\ 1+2x+x &= 1+x \\ 2x &= 0.\end{aligned}$$

□

Proof. Suppose $x \in A$ represents the class $\bar{x} \in A/\mathfrak{p}$. Then,

$$\bar{x}^2 = (\mathfrak{p}+x)(\mathfrak{p}+x) = \mathfrak{p}+x^2 = \mathfrak{p}+x = \bar{x}.$$

Since $\mathfrak{p}$ is prime, $A/\mathfrak{p}$ is an integral domain. Hence, $\bar{x}^2 = \bar{x} \implies \bar{x}(\bar{x}-1) = 0 \implies \bar{x} \in \{0,1\}$. Thus, $A/\mathfrak{p}$ is the field of size 2, making $\mathfrak{p}$ a maximal ideal. □

1.12 Exercise - Complete

Proof. If x is a unit, $x^2 = x \implies x^2x^{-1} = xx^{-1} \implies x = 1$. If x is not a unit, then $x \in \mathfrak{m}$. That means that $1-x$ is not in $\mathfrak{m}$ (otherwise $x+(1-x) = 1 \in \mathfrak{m}$) and consequently that $1-x$ is a unit. $x^2 = x \implies x(1-x) = 0$, but since $1-x$ is a unit, we have $x = 0$. □

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